Cisco Secure Workload User Education Guide

Introduction, Video Library, and Onboarding Runbook

Cisco Secure Workload User Education

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Purpose

This guide helps users understand **Cisco Secure Workload (CSW)**, use a curated video library for learning, and follow a practical onboarding runbook for discovery, deployment, application dependency mapping, policy modeling, and staged micro-segmentation enforcement.

Use this as a user-friendly starting point. It is intentionally written in practical language, not as a replacement for Cisco product documentation, release notes, or design validation.

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1. What Cisco Secure Workload Really Is

Cisco Secure Workload is a workload visibility and micro-segmentation platform. It helps customers understand how applications communicate, convert that understanding into policy, and enforce least-privilege communication across servers, virtual machines, cloud workloads, and supported application environments.

In simple terms, CSW answers four practical questions:

1. **What do I have?**
   Inventory workloads, labels, operating systems, processes, and available metadata from agents, cloud connectors, CMDB systems, and other integrations.

2. **What talks to what?**
   Observe L3 / L4 flows and, where supported, process-level context so teams can understand real application dependencies instead of guessing.

3. **What should be allowed?**
   Use observed flows, labels, application groups, and policy modeling to recommend or build least-privilege segmentation policies.

4. **How do I enforce safely?**
   Move from visibility to monitor mode, then to staged enforcement, with rollback and operational validation.

CSW is not just a firewall rule tool. It is an application dependency mapping, workload labeling, policy modeling, and micro-segmentation platform. The value comes from combining telemetry, labels, policy simulation, and controlled enforcement.

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2. What CSW Solves: Lateral Movement and Ransomware

Cisco Secure Workload exists to break the assumption that an attacker who gets inside the network can move freely. Most enterprise networks are still relatively flat at the workload layer — once one server is compromised through a phishing link, an exposed admin port, a vulnerable web app, or a stolen credential, the attacker can usually reach hundreds of other servers, file shares, and databases over the same internal network paths that legitimate applications use.

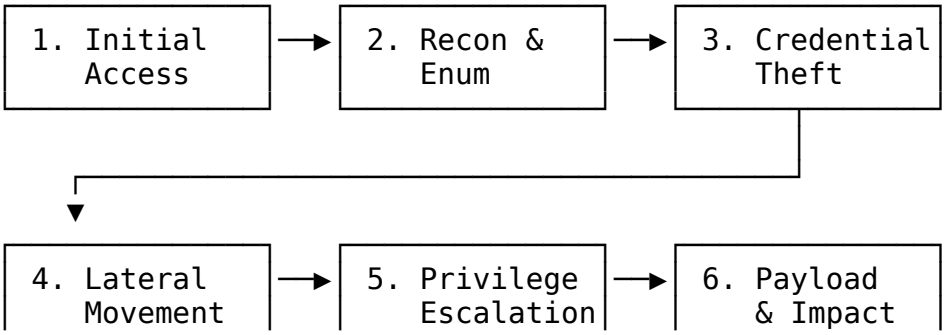
CSW reduces the **blast radius** of an intrusion by limiting each workload to **only the communication it actually needs**, based on real observed application behavior.

2.1 The Lateral Movement Problem

After initial access, modern attackers and most ransomware operators follow a predictable pattern:

1. **Land on one workload** — phishing, exposed RDP / SSH, exploited service, supply chain.
2. **Discover the internal network** — port scans, SMB enumeration, Active Directory reconnaissance.
3. **Steal credentials** — Mimikatz, LSASS dumps, Kerberoasting.
4. **Move laterally** — RDP, SMB, WinRM, WMI, PsExec, SSH, RPC, and other lateral admin tools.
5. **Escalate to high-value targets** — domain controllers, backup servers, file servers, databases, hypervisors.
6. **Stage and detonate the payload** — ransomware, data exfiltration, destruction.

Visually, the kill chain looks like this:



```
└──────────────┘    └──────────────┘    └──────────────┘

  * CSW intervenes between steps 4 and 5:
    least-privilege policy removes the
    workload-to-workload network paths
    these steps depend on. The attacker
    can still land on host 1, but cannot
    reach hosts 2..N from there.
```

Steps 2 through 5 **all depend on the network allowing workload-to-workload traffic that no business application actually requires**. That is exactly the layer CSW controls.

2.2 How CSW Stops or Slows Ransomware

Ransomware groups make money by encrypting many systems in a short time window. They depend on:

- Unrestricted **SMB / file share access** between workstations and servers.
- Open **RDP / WinRM / WMI / PsExec** paths between servers.
- Reachable **backup servers, file servers, and hypervisors** from compromised endpoints.
- Free **east-west traffic** that lets the malware fan out before defenders can react.

CSW directly attacks each of those preconditions:

Ransomware behavior	What CSW does about it
SMB / RDP / WinRM fan-out from a compromised workload	Allow only application-required ports between specific workload groups; deny lateral admin protocols by default.
Mass file-share encryption	Restrict file-server access to the specific workloads, users, and processes that genuinely need it.
Reaching backup or recovery systems	Place backup servers in their own scope; permit traffic only from backup agents, not from general workloads.
Reaching domain controllers or identity tier	Scope the identity / Active Directory tier so only required clients and admin jump hosts can reach it.
Hopping between application tiers	Enforce per-app and per-tier policy (web → app → database) so a compromised web tier cannot reach unrelated databases.
Spreading from dev / test / lab into prod	Hard-segment prod from non-prod so a non-prod compromise cannot pivot into prod.

Ransomware behavior	What CSW does about it
Long undetected dwell time	Surface every blocked or anomalous flow as evidence for SOC and incident response.

The result is that **ransomware that lands on one workload finds the network around it almost empty**: the protocols it needs to spread are blocked by policy, and every attempt is logged.

2.3 Why We Need It

- **Flat networks no longer match the threat model.** Perimeter firewalls do not stop an attacker who is already inside.
- **Identity-based and EDR controls are necessary but not sufficient.** They catch behavior on the host. CSW removes the network paths the attacker would use between hosts.
- **Crown-jewel applications need explicit protection.** Payments, claims, customer data, intellectual property, and backup infrastructure should not be reachable from a random user workstation or a low-tier dev server.
- **Compliance and audit demand it.** PCI, HIPAA, SOX, and most internal security frameworks expect documented segmentation between regulated and non-regulated systems.
- **It must not break applications.** CSW's discovery-first model (map dependencies → label workloads → model policy → enforce in stages) is what makes segmentation finally feasible in real enterprises.

In one sentence: **CSW exists so that when — not if — one workload is compromised, the attacker cannot reach the next one.**

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3. Why Customers Care

Customers usually evaluate CSW because they need to reduce lateral movement risk without breaking applications. Traditional network segmentation often depends on VLANs, subnets, firewall zones, or static IP lists. That model becomes difficult when applications span data centers, public cloud, containers, and shared services.

CSW helps customers:

- Discover application dependencies before enforcing policy.
- Build policies based on labels and application intent, not only IP addresses.
- Protect high-value or vulnerable applications with tighter controls.
- Align policy with CMDB data, cloud tags, application ownership, and observed behavior.
- Roll out segmentation in phases, starting with visibility and policy modeling.
- Produce evidence for security, audit, and architecture teams.

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4. Core CSW Concepts for Users

Concept	User-friendly explanation
Workload	A server, VM, cloud instance, container host, or supported compute asset being observed or protected.
Agent / enforcer	Software installed on supported workloads to collect telemetry and, when enabled, enforce policy.
Flow visibility	The observed network communication between workloads, usually source, destination, protocol, port, and time.
Process visibility	Mapping a network flow to the local process or service using it, where supported.
Labels	Metadata used to describe workloads, such as application, environment, role, owner, location, or risk.
Scope	A logical boundary for organizing workloads and policies.
Policy modeling	Previewing what would be allowed or blocked before enforcing.
Monitor mode	Observe and report policy behavior without blocking traffic.
Enforcement mode	Actively allow or deny traffic based on policy.
Connectors	Integrations that bring in context from systems such as ServiceNow, cloud providers, or other sources.
Edge appliance	A CSW virtual appliance used to run certain connectors, such as the ServiceNow connector.
Ingest	A pipeline or appliance pattern used where external telemetry ingestion is required. Validate design with Cisco for the customer use case.

5. CSW Video Library

The video catalog below mirrors the Cisco Secure Workload learning material. Sub-section 5.1 contains the ten foundational training videos from the source Cisco slide. Sub-sections 5.2 through 5.7 extend the catalog with deeper topics commonly requested in customer conversations: security and risk reduction, segmentation use cases, firewall integration, F5 /

BIG-IP integration, identity and DNS integrations, and containers. All entries link to YouTube unless explicitly noted otherwise.

5.1 Core CSW Training

#	Video	Description	Link
1	Cisco Secure Workload: Agent Configuration Profile	Overview of configuring agent profiles to manage workload agents efficiently for security enforcement.	Watch
2	Cisco Secure Workload: Agent Configuration and Deployment – Golden Image VDI	Demonstrates deploying Secure Workload agents in virtual desktop infrastructure environments using golden images.	Watch
3	Cisco Secure Workload: Scopes	Explains how to use scopes to group workloads logically for policy application and management.	Watch
4	Cisco Secure Workload: Labels	Details the use of labels to tag workloads for granular policy enforcement and visibility.	Watch
5	Cisco Secure Workload: Inventory Filters	Shows how to filter inventory to focus on specific workloads or groups for analysis and policy.	Watch
6	Cisco Secure Workload: Production and Test Risk Reduction	Covers strategies to reduce risk in production and test environments using Secure Workload	Watch

#	Video	Description	Link
		policies.	
7	Cisco Secure Workload: Application Dependency Mapping & Policy Analysis	Demonstrates mapping application dependencies and analyzing policies for effective microsegmentation.	Watch
8	Cisco Secure Workload: Policy Visual and Quick Analysis	Provides tools for visualizing and quickly analyzing security policies and their impact.	Watch
9	Cisco Secure Workload: Dynamic Workloads & Policy	Explains handling dynamic workloads and adapting policies accordingly.	Watch
10	Cisco Secure Workload: Flow Analysis	Details flow analysis capabilities to understand traffic patterns and detect anomalies.	Watch

5.2 Security and Risk Reduction

#	Video	Description	Link
11	Cisco Secure Workload: Vulnerabilities and Risk Reduction	Covers using Secure Workload to identify vulnerable workloads, prioritize risk, and reduce exposure.	Watch
12	Cisco Secure Workload: Security Dashboard	Walks through the Secure Workload security dashboard: posture, risk indicators, and	Watch

#	Video	Description	Link
		drill-downs.	
13	Cisco Secure Workload: Forensics	Demonstrates Secure Workload's forensics capability for capturing process and flow evidence during investigations.	Watch
14	Cisco Secure Workload: Log4J Risk Reduction	Shows how to use Secure Workload to identify and contain Log4J (Log4Shell) exposure across workloads.	Watch
15	Cisco Secure Workload: SSH Risk Reduction	Demonstrates segmenting and controlling SSH paths between workloads using Secure Workload policy.	Watch

5.3 Segmentation Use Cases

#	Video	Description	Link
16	Cisco Secure Workload: Terminal Services Segmentation	Covers segmenting Microsoft Terminal Services / Remote Desktop Services environments with Secure Workload.	Watch
17	Cisco Secure Workload: Virtual Desktop Interface (VDI) Segmentation	Covers segmenting VDI environments where many users share a small set of golden images.	Watch

5.4 Secure Workload + Secure Firewall Integration

#	Resource	Description	Link
18	Secure Workload & Firewall Integration (Part 1)	Part 1 of the integration series: introduction, design, and architecture.	Watch
19	Secure Workload & Firewall Integration (Part 2)	Part 2: deployment patterns and policy flow.	Watch
20	Secure Workload & Firewall Integration (Part 3)	Part 3: enforcement, telemetry, and operational considerations.	Watch
21	Cisco Secure Workload and Secure Firewall White Paper	Public Cisco reference paper covering the joint architecture and design principles.	Cisco.com
22	Secure Workload & Firewall Integration Deep Dive	Detailed reference on integration design principles, architecture, and use-cases.	Read

5.5 F5 / BIG-IP Integration

#	Video	Description	Link
23	F5 BIG-IP and Cisco Tetration: APM Visibility	Shows how Secure Workload (formerly Tetration) ingests F5 BIG-IP APM data for application visibility.	Watch
24	Cisco Tetration and F5 BIG-IP AFM	Covers integration with F5 BIG-IP AFM (Advanced Firewall Manager) for	Watch

#	Video	Description	Link
		additional flow context.	
25	Cisco Tetration and F5 BIG-IP IPFIX Configuration	Configuration walk-through for sending IPFIX flow records from F5 BIG-IP into Secure Workload.	Watch

5.6 Identity, DNS, and Other Integrations

#	Video	Description	Link
26	Cisco Secure Workload — DNS Server Integration	Covers integrating DNS server context so workload-to-name mappings improve flow attribution and policy.	Watch
27	Cisco Secure Workload — Infoblox Integration	Covers integrating Infoblox DNS / IPAM data for richer workload context inside Secure Workload.	Watch
28	Cisco Secure Workload and Algosec Integration	Covers integrating Algosec for policy analysis and firewall-policy lifecycle alongside Secure Workload.	Watch
29	Cisco Secure Workload and ISE (In Action)	Demonstrates Secure Workload integration with Cisco ISE for user and device identity context.	Watch
30	FMC Integration with Edge / Ingest / Appliance	Covers integrating Cisco FMC (Firewall	Watch

#	Video	Description	Link
		Management Center) through Secure Workload's Edge, Ingest, and Appliance integration paths.	

5.7 Containers and Kubernetes

#	Video	Description	Link
31	Cisco Secure Workload: Agent K8s	Demonstrates the Secure Workload agent in Kubernetes / containerized environments.	Watch

Suggested Watch Order

The source Cisco slide does not prescribe a viewing order. For a new user, the progression below moves from foundations to deployment to policy and analytics, then opens up by use case and integration:

1. **Foundations** — Scopes (#3), Labels (#4), Inventory Filters (#5).
2. **Agent rollout** — Agent Configuration Profile (#1), then Agent Configuration and Deployment – Golden Image VDI (#2) if VDI is in scope.
3. **Core value: visibility → policy** — Application Dependency Mapping and Policy Analysis (#7), Flow Analysis (#10), Dynamic Workloads and Policy (#9), Policy Visual and Quick Analysis (#8), Production and Test Risk Reduction (#6).
4. **Security and risk** — Security Dashboard (#12), Vulnerabilities and Risk Reduction (#11), Forensics (#13), Log4J Risk Reduction (#14), SSH Risk Reduction (#15).
5. **Specific segmentation use cases** — Terminal Services (#16), VDI Segmentation (#17).
6. **Integrations as needed by the deal / POV** — Firewall (#18–#22), F5 / BIG-IP (#23–#25), DNS / Infoblox (#26, #27), Algosec (#28), ISE (#29), FMC (#30).
7. **Containers** — Agent K8s (#31) when Kubernetes is in scope.

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6. How to Explain CSW in a Customer Meeting

30-second version

Cisco Secure Workload helps customers discover how applications communicate and then safely enforce least-privilege segmentation policies around those applications. It starts with visibility and dependency mapping, uses labels and observed behavior to create policy, and supports staged enforcement so teams can reduce lateral movement risk without breaking applications.

2-minute version

Most enterprises want micro-segmentation, but they hesitate because they do not fully know what their applications depend on. CSW starts by collecting workload telemetry, flow visibility, process context, cloud metadata, and CMDB labels. That data helps application teams understand real dependencies. From there, teams can model policy, see what would be blocked, tune the rules, and move selected workloads from monitor mode to enforcement. CSW is especially useful when customers need consistent segmentation across data center and cloud environments.

Common positioning

- **For security:** Reduce lateral movement and contain high-risk applications.
- **For network teams:** Shift from static IP / subnet rules to workload-aware policy.
- **For application teams:** Discover dependencies before enforcement and avoid unexpected outages.
- **For cloud teams:** Use cloud tags and dynamic metadata so policy follows workloads.
- **For operations:** Start in monitor mode, model changes, and roll out enforcement gradually.

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7. CSW Onboarding Runbook

Phase 0 - Confirm Scope and Outcomes

Goal: Agree on what the customer wants to prove.

Steps:

1. Identify the target customer environment: data center, public cloud, containers, or a mix.
2. Select 2-3 representative applications.
3. Pick at least one low-risk application and one higher-value application.
4. Define success criteria:
    - Flow visibility demonstrated.
    - Process visibility demonstrated.
    - Application dependency map reviewed by app owner.
    - Labels imported or created.
    - Policy modeled before enforcement.
    - One controlled enforcement test completed.

Output: POV scope, application list, stakeholder list, and success criteria.

Phase 1 - Prepare the Tenant and Access

Goal: Prepare the CSW management environment and user access.

Steps:

1. Confirm CSW tenant or cluster access.
2. Create required administrator, operator, and viewer accounts.
3. Configure SSO / MFA / RBAC if in scope.
4. Confirm API access if automation or reporting is part of the POV.
5. Document who can create labels, scopes, policies, and enforcement changes.

Output: Access matrix, RBAC model, and tenant readiness confirmation.

Phase 2 - Build the Label Strategy

Goal: Establish labels before policy design.

Recommended starting labels:

Label type	Example values	Source
Application	`payments`, `claims`, `customer-portal`	CMDB, app owner, manual
Environment	`prod`, `non-prod`, `dev`, `test`	CMDB, cloud tag
Role	`web`, `app`, `db`, `middleware`	CMDB, observed behavior
Location	`dc1`, `dc2`, `aws`, `azure`	Cloud connector, manual
Owner	`team-a`, `team-b`	CMDB
Risk	`high`, `medium`, `low`	vulnerability scanner, security team

Steps:

1. Agree on a minimal label taxonomy.
2. Map sources for each label: ServiceNow, cloud tags, manual, API, or observed behavior.
3. Avoid creating too many labels before discovery. Start simple.
4. Validate that labels can support the first policy examples.

Output: Initial label dictionary and source-of-truth mapping.

Phase 3 - Deploy Agents and Connectors

Goal: Start collecting telemetry and context.

Steps:

1. Install CSW agents on selected Linux / Windows workloads.

2. Confirm agent health and visibility.
3. Deploy a CSW Edge virtual appliance if connector workloads are required.
4. Configure ServiceNow connector if CMDB labels are in scope.
5. Configure cloud connectors if public cloud inventory / tags are in scope.
6. Configure Ingest only if external flow or metadata ingestion is required and validated by Cisco design.

Output: Healthy agents, connector status, imported labels, and visible workload inventory.

Phase 4 - Discover Application Dependencies

Goal: Learn how the application actually communicates.

Steps:

1. Let workloads run through normal business transactions.
2. Capture L3 / L4 flow data.
3. Review process-level visibility where supported.
4. Build application dependency maps.
5. Review dependencies with application owners.
6. Identify unexpected flows, shared services, admin paths, and risky protocols.

Output: Dependency map, flow summary, process evidence, and app-owner review notes.

Phase 5 - Model Policy

Goal: Build policy without breaking the application.

Steps:

1. Use observed flows and labels to create recommended allow rules.
2. Group policy by application tiers where possible.
3. Identify required shared services such as DNS, NTP, authentication, logging, monitoring, backup, and patching.
4. Model the policy before enforcement.
5. Review predicted blocked flows with application and operations teams.
6. Tune policy until the expected business transactions are allowed.

Output: Proposed policy, modeled impact, predicted denies, and approval to test enforcement.

Phase 6 - Start in Monitor Mode

Goal: Validate policy behavior before blocking.

Steps:

1. Apply policy in monitor or non-blocking mode where supported.
2. Run application test transactions.
3. Review policy violations and unexpected flows.
4. Update labels or policy rules as needed.
5. Confirm application owners are comfortable with the modeled impact.

Output: Monitor-mode results, policy tuning notes, and enforcement readiness decision.

Phase 7 - Enforce in a Controlled Scope

Goal: Prove micro-segmentation safely.

Steps:

1. Select a small enforcement scope.
2. Schedule a change window.
3. Confirm rollback plan.
4. Move selected workloads to enforcement.
5. Test allowed traffic.
6. Test intentionally denied traffic.
7. Confirm logs, alarms, and reports show the expected result.
8. Monitor application health after enforcement.

Output: Enforcement proof, blocked-flow evidence, allowed-flow evidence, rollback plan, and operational signoff.

Phase 8 - Operationalize

Goal: Make CSW sustainable after the POV.

Steps:

1. Define ownership for labels, scopes, and policies.
2. Document policy change process.
3. Integrate logs / alarms with SIEM or SOAR if in scope.
4. Create recurring policy review cadence.
5. Build API or CI/CD automation where needed.
6. Define exception handling for unsupported workloads.
7. Expand from the initial application set to the next wave.

Output: Operating model, change process, exception process, and expansion roadmap.

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8. Discovery Questions

Use these questions before sizing or designing a POV:

1. Which business application should be used for the first dependency map?
2. Which application owner can validate the map?
3. Which environments are in scope: prod, non-prod, dev, test?
4. Which platforms are in scope: VMware, bare metal, public cloud, containers?
5. Which operating systems are in scope?
6. Is ServiceNow the CMDB source of truth?
7. Are cloud tags clean enough to use as labels?
8. Is MFA / SSO required for CSW administrator access?
9. Are agents allowed on the selected workloads?
10. Is enforcement part of the POV, or visibility-only?
11. What is the rollback process for policy enforcement?
12. Which SIEM / SOAR tools should receive alarms or logs?
13. What vulnerability scanner or risk source should be used for high-risk labels?
14. Who approves policy changes?
15. What evidence does the customer need at the end of the POV?

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9. POV Evidence Checklist

Evidence	Why it matters
Agent health screenshot	Proves telemetry source is active
Workload inventory screenshot	Proves assets are visible
Label inventory screenshot	Proves metadata is usable
ServiceNow connector status	Proves CMDB integration path
Cloud connector status	Proves cloud context ingestion
Flow table export	Proves L3 / L4 visibility
Process-level flow screenshot	Proves process context where supported
Application dependency map	Proves discovery value
Policy recommendation	Proves policy can be derived from observed behavior
Policy modeling output	Proves change impact can be reviewed before enforcement
Monitor-mode policy result	Proves safe pre-enforcement workflow
Enforcement blocked-flow test	Proves actual segmentation control
Allowed business transaction test	Proves enforcement did not break required traffic
Audit log	Proves administrative traceability

Evidence	Why it matters
API output	Proves automation path
SIEM event	Proves operational integration

10. Common Pitfalls

Pitfall	How to avoid it
Starting with enforcement too early	Start with discovery, labels, and modeling before blocking traffic.
Using too many labels too soon	Start with application, environment, role, location, owner, and risk. Expand later.
Skipping application owner validation	Dependency maps must be reviewed by the app owner before policy.
Ignoring shared services	Always account for DNS, NTP, identity, logging, monitoring, backup, and patching.
Treating cloud and data center separately	Use consistent labels across both so policy is portable.
Forgetting rollback	Every enforcement test needs a rollback plan and owner.
Overclaiming integration support	Confirm exact platform and version support with current Cisco documentation.

11. Simple CSW Talk Track

Use this flow in customer conversations:

1. **Start with the risk:** Attackers move laterally after initial compromise. Flat networks make that easier.
2. **Explain the blocker:** Customers hesitate to segment because they do not know every application dependency.
3. **Position CSW:** CSW discovers dependencies first, then models policy, then enforces gradually.
4. **Make it practical:** We start with a few representative apps, import labels, review flows, model policy, and only then enforce.
5. **Close with outcome:** The customer gets application maps, policy recommendations, enforcement proof, and an expansion roadmap.

12. Open Items Before Publishing Final Version

1. Replace all `URL needed` placeholders in the video library with real video links.
2. Confirm whether this repo should be public, private, or internal-only.
3. Confirm whether customer-specific examples should be removed or kept generic.
4. Add official Cisco documentation links for the final user education version.
5. Validate current release-specific support matrices before presenting platform coverage.